EXPERIMENT 16

SIMULATION AND ANALYSIS OF UNIFORM AND NON-UNIFORM QUANTIZATION USING MATLAB

```
clc;
clear;
close all;
% Input Signal t = 0:0.001:1;
x = sin(2*pi*5*t); % Uniform Quantization (8 Levels) L = 8;
x_uni = round((x+1)*(L-1)/2);
x_uni = (x_uni*2/(L-1))-1;
% Non-Uniform Quantization (μ-law) mu = 255;
x_comp = sign(x).*log(1+mu*abs(x))/log(1+mu);
x_non = round((x_comp+1)*(L-1)/2);
x_non = (x_non*2/(L-1))-1;
% Plot Results figure;
subplot(3,1,1) plot(t,x) ;
title('Original Signal') ;
grid on ;
subplot(3,1,2) ;
stairs(t,x_uni) ;
title('Uniform Quantization') ;
```

```
grid on ;

 subplot(3,1,3) ;

stairs(t,x_non) ;

 title('Non-Uniform Quantization (\mu-law)') ;

grid on ;
```

